

Improved Automatic Colorization by Optimal Pre-colorization

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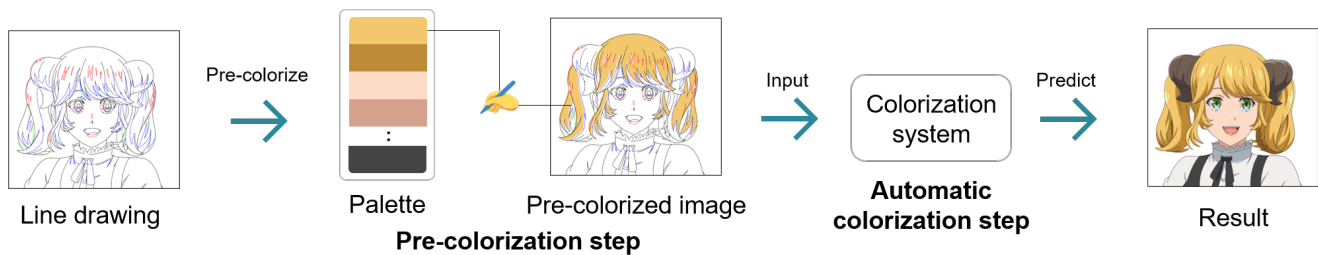
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Figure 1: The proposed colorization pipeline

ABSTRACT

Automatic line-drawings colorization of anime characters is a challenging problem in computer graphics. The previous fully automatic colorization method suffers from colorization accuracy and costs colorization artists to validate and correct colorization errors. We propose to improve the colorization accuracy by introducing “pre-colorization” step into our production pipeline that requests user to manually colorize partial regions in line-drawings before doing automatic colorization. The pre-colored regions work as clues to colorize the other regions and improve the colorization accuracy. We found an optimal region to be pre-colored to obtain the best automatic colorization performance.

CCS CONCEPTS

• Computing methodologies → Image manipulation.

KEYWORDS

datasets, automatic colorization, anime production, pre-colorization

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1 INTRODUCTION

In anime production pipeline, colorizing line-drawings is a time-consuming task. The accuracy of a conventional fully automatic colorization method is not still sufficient for practical use. Therefore, it tends to require additional costs to validate and correct colorization errors, manually. Ishii et al. [2020] proposed a method of automatic colorization based on prediction confidence to reduce incorrect predictions. This study therefore aims to improve the colorization accuracy by using partially colorized line-drawings as an input image. Also, we propose an optimal region to be colorized manually to bring higher estimation accuracy.

2 PROPOSED METHOD

Figure 1 shows an overview of the proposed method. The proposed method consists of pre-colorization and automatic colorization steps. Firstly, pre-colorization step manually colorizes a region on a line-drawing from a given color palette, then creates a pre-colored image. Secondly, the next step automatically colorizes the partially-colored image with a colorization system of [Ishii et al. 2020] and receives a fully-colored result (automatic colorization step). The pre-colored region can be a clue to improve the colorization accuracy. Thus, this pre-colorization step is expected to reduce overall costs, including correcting errors after the system’s colorization.

3 EXPERIMENT AND RESULTS

Dataset and Pre-colorization. In this research, we focus on one specific character shown in Figure2 for the experiment. Also, six

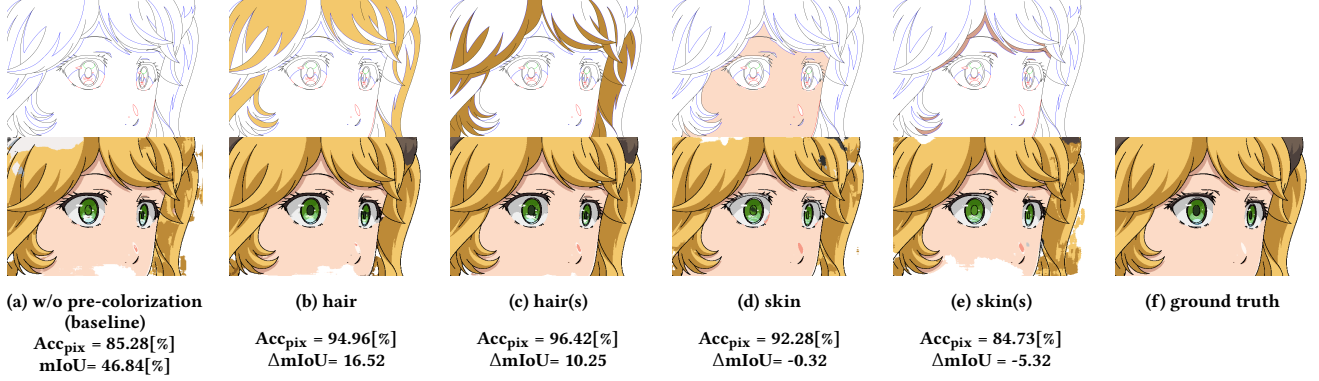


Figure 2: Colorization results (s):(shadow) ©Junpei Inuzuka, SHUFUNOTOMO INFOS/Restaurant to Another World 2 Project

Table 1: Quantitative comparison with Acc_{pix} and mIoU for the results of each trained model. (s): (shadow)

Regions	$\text{Acc}_{\text{pix}}[\%]$	ΔmIoU	Pre-colored regions[%]
hair	97.37	4.51	6.10
hair(s)	97.35	0.51	3.42
skin	96.49	2.69	3.94
skin(s)	96.03	-0.57	0.73

major regions of the character are selected as candidates to be manually colorized. A dataset consists of 936 pairs of pre- and post-colored line-drawings of a character from actually broadcast TV anime series shown in figure 2. This dataset is prepared for each of six candidates. Each dataset consists of partially colorized line-drawings of the corresponding region for inputs, and fully colorized images for outputs. We split the dataset into three subsets; 743 for training, 94 for validation, and 99 for testing.

Method. The purpose of this experiment is to find an optimal region from six candidate regions to be manually colorized. First, we trained each six U-nets[Ronneberger et al. 2015] by using each of six pre-colored training datasets under the same hyperparameters in 300 epochs. Additionally, we also trained the same network using the original, un-colored version of the training dataset in the same manner as a baseline for comparison. Second, the line-drawings in the test data were partially colorized, and the colorizing prediction was tested for each selected color. Finally, we calculate mean Intersection over Union (mIoU) and pixel-wise accuracy (Acc_{pix}). For fair comparison, we do not use any predicted results of the pre-colored to compute mIoUs of our method.

Results and discussion. Figure 2 shows the colorization results. The top and bottom rows show the input and output images of the colorization system, respectively. Compared with the colorized result which is obtained by input line-drawing without any pre-colorization (a), the estimated pixel accuracy of the results obtained by partially pre-colored line-drawings (b)-(e) are higher in most cases. More specifically, pre-colored *hair* region (c) brings higher accuracy than *hair(shadow)*, *skin*, and *skin(shadow)*. The mIoUs of these estimated images indicate similar results. Note, we only

display four of six candidates' results (b)-(e). See the supplemental PDF for all candidates. Figure 2 suggests that pre-colorizing larger areas would improve pixel accuracy and mIoUs. To validate this, table 1 shows the relation between pixel accuracy, ΔmIoU , and the area of pre-colored regions. ΔmIoU is a difference between mIoU of pre-colorization method and baseline's mIoU, which represents the improved accuracy of the proposed method for the un-colored area compare with the baseline. Table 1 shows that the area with the largest difference, i.e., the area that contributed to the improvement in accuracy, is the *hair* area. The colorization results suggest that, at present, pre-colorization of relatively large-area areas tends to improve accuracy. In addition to these results, the model training time was reduced to two-thirds of the time required to achieve the same level of accuracy as the previous method.

4 CONCLUSION

In this research, we found that the larger regions tend to be optimal to be manually pre-colored for the improvement of the automatic colorization accuracy. As a preliminary study, we only focus on one specific character, however, it is necessary to generalize the result by applying our technique to various characters which is a future work of our research. Furthermore, an interactive system that repeats automatic colorization and manual correction will be used on a trial basis at production sites to investigate its impact on overall cost reduction.

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